

Assignment 6: At-Risk Patient Logic Report- Ansarah Mohammed

Vital Sign Chosen: Mean Arterial Pressure (MAP)

Purpose of the Algorithm:

For Assignment 6, I chose Mean Arterial Pressure (MAP) as the vital sign used to flag patients who may be “at-risk.” As the clinical background of MAP, including its definition, formula, and triage relevance, was addressed in Assignment 4, this report focuses specifically on developing and defending the at-risk patient logic.

The purpose of this algorithm is to identify patients whose MAP values may need further review. In my Week 0 cleaning work, I used 40–180 mmHg as the valid range for MAP. However, a value being valid in the dataset does not automatically mean it is safe for the patient. Because of this, my logic separates invalid data values from clinically concerning values.

Algorithm Design and Threshold Justification:

For this algorithm, I used the following MAP categories:

- Missing MAP value: Needs Review
- $\text{MAP} < 40 \text{ mmHg}$ or $\text{MAP} > 180 \text{ mmHg}$: Invalid MAP Value
- $\text{MAP} < 65 \text{ mmHg}$: At-Risk: Low MAP
- $\text{MAP} > 110 \text{ mmHg}$: At-Risk: High MAP
- MAP between 65 and 110 mmHg: Not At-Risk Based on MAP

I used $\text{MAP} < 65 \text{ mmHg}$ as the low-risk threshold because low MAP may suggest poor organ perfusion. This means that important organs such as the brain, heart, and kidneys may not be receiving enough blood and oxygen. Even though values between 40 and 65 mmHg are still within the valid dataset range, they should not be treated as safe.

I used $\text{MAP} > 110 \text{ mmHg}$ as the high-risk threshold because very high MAP may place extra strain on the heart and blood vessels. A value above 110 mmHg may still be valid data, but it should be flagged so that the patient can be reviewed properly.

This approach is important because health-related values should not be treated as simple outliers only. Some unusual values may represent real clinical concerns and may affect a patient’s safety.

Pseudocode:

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INPUT: Patient MAP value

IF MAP value is missing THEN

    Patient risk status = "Needs Review"

ELSE IF MAP < 40 OR MAP > 180 THEN

    Patient risk status = "Invalid MAP Value"

ELSE IF MAP < 65 THEN

    Patient risk status = "At-Risk: Low MAP"

ELSE IF MAP > 110 THEN

    Patient risk status = "At-Risk: High MAP"

ELSE

    Patient risk status = "Not At-Risk Based on MAP"

OUTPUT: Patient risk status
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Explanation of the Logic:

The algorithm first checks if the MAP value is missing. If it is missing, the patient is marked as “Needs Review” because missing clinical data should not be assumed to be normal.

Next, the algorithm checks whether the MAP value is outside the valid range of 40–180 mmHg. If it is outside this range, it is labelled as an “Invalid MAP Value” because it may be unrealistic or incorrectly entered.

After that, the algorithm checks for clinically unsafe MAP values. If the MAP is below 65 mmHg, the patient is flagged as “At-Risk: Low MAP.” If the MAP is above 110 mmHg, the patient is flagged as “At-Risk: High MAP.” If the MAP is between 65 and 110 mmHg, the patient is labelled as “Not At-Risk Based on MAP.” This order makes the logic clear because it checks for missing data first, then invalid data, and then patient risk.

Link to Triage Decisions:

Although this algorithm uses MAP, real triage decisions should not depend on one vital sign only. Other vital signs such as pulse rate, respiratory rate, temperature, oxygen saturation, systolic blood pressure, and level of consciousness should also be considered.

For example, a patient may have a normal MAP but still be at-risk if they have low oxygen saturation, a very high pulse, fever, or confusion. Therefore, this MAP algorithm should be used as a screening tool to highlight patients who may need further review, not as a final diagnosis.

Conclusion:

In conclusion, this algorithm uses MAP to flag patients who may need further clinical review. It checks for missing values, invalid values, low MAP, and high MAP. The main idea is that a MAP value can be valid in the dataset but is still unsafe for the patient. By flagging MAP values below 65 mmHg and above 110 mmHg, the algorithm provides a simple and practical way to identify patients who may be at-risk.

References:

- DeMers, D., & Wachs, D. (2023). Physiology, mean arterial pressure. In StatPearls. StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK538226/>
- EBSCO. (2024). Mean arterial pressure (MAP). EBSCO Research Starters. <https://www.ebsco.com/research-starters/health-and-medicine/mean-arterial-pressure-map>